# Robot Design Corrections

After building the first prototype of the robot and beginning initial testing, we discovered that relying entirely on distance sensors for continuous wall tracking was inefficient and difficult to execute consistently. The sensor readings varied depending on the robot's angle relative to the wall, which reduced navigation stability.

Because of this, we changed our navigation strategy. Instead of using the distance sensors for constant wall following, we decided to use more reliable positional metrics and internal movement calculations for controlling the robot's movement. The distance sensors are now primarily used to help detect when the robot is approaching a colored obstacle and when turning maneuvers should begin.

During the early design phase, our original plan was to build a custom 3D-printed chassis. An early custom 3D-printed chassis design was created during development. However, after evaluating durability and ease of modification, the team transitioned to a double acrylic frame design. Although a complete 3D model was designed, we ultimately decided not to use it because of concerns regarding durability, reliability, and manufacturing time. The acrylic frame provided greater structural stability and was easier to modify during testing.

While assembling the robot, we also encountered several wiring-related issues. Some motors were initially connected to incorrect pins, which caused unpredictable robot movement and inconsistent motor behavior. To solve this problem, the robot had to be partially disassembled and rewired correctly. This process also helped improve our understanding of cable organization and electronics integration.

These early problems and corrections significantly improved the overall reliability of the robot and helped guide later development decisions.